

V-plan AXI-VIP

Description of AXI

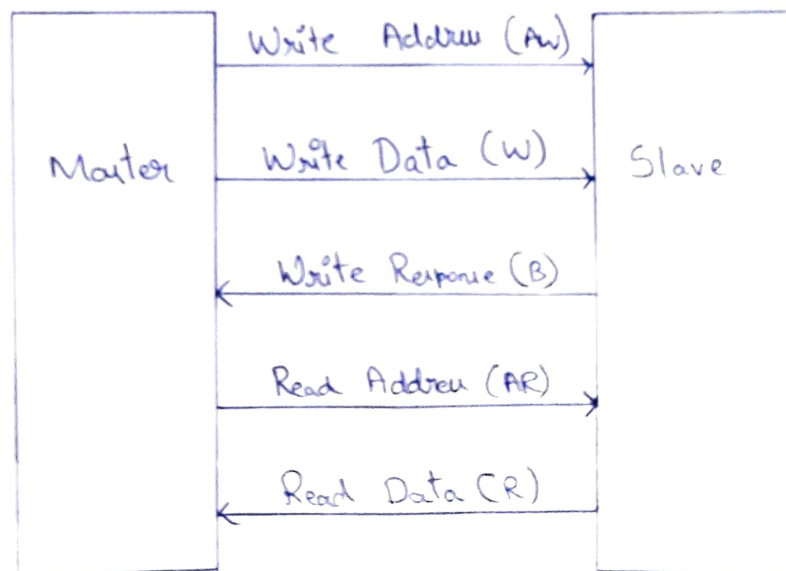
AXI is a protocol of the AMBA family. Advanced Extensible Interface has been used as a protocol for communication b/w processor & memory units in advanced microcontrollers. Also in SoCs of various IPs and smartphones.

AXI is a burst type of protocol, which means, there exists a series of data transfers for a single transaction.

Transactions are of three types 1) Fixed 2) Increment and 3) Wrap

These transactions are supported by means of 5 channels, namely Write Address (AW), Write Data (W), Write Response (B), Read Address (AR) and Read Data (R)

Block diagram



Channels And their Signals

1. Write Address

Name	Source	Description
AWVALID	Master (M)	Valid Indicator
AWREADY	Slave (S)	Ready Indicator
AWID	M	Transaction Identifier
AWADDR	M	Transaction Address
AWLEN	M	Transaction Length
AWSIZE	M	Transaction Size
AWBURST	M	Transaction type

2. Write data

WVALID	M	Valid
WREADY	S	Ready
WID	M	Identifier
WDATA	M	Transfer data
WSTRB	M	Byte lane data

3.

Write response

BVALID	S	Valid
BREADY	M	Ready
BID	S	Identifier
BRESP	S	Write response

4. Read Address

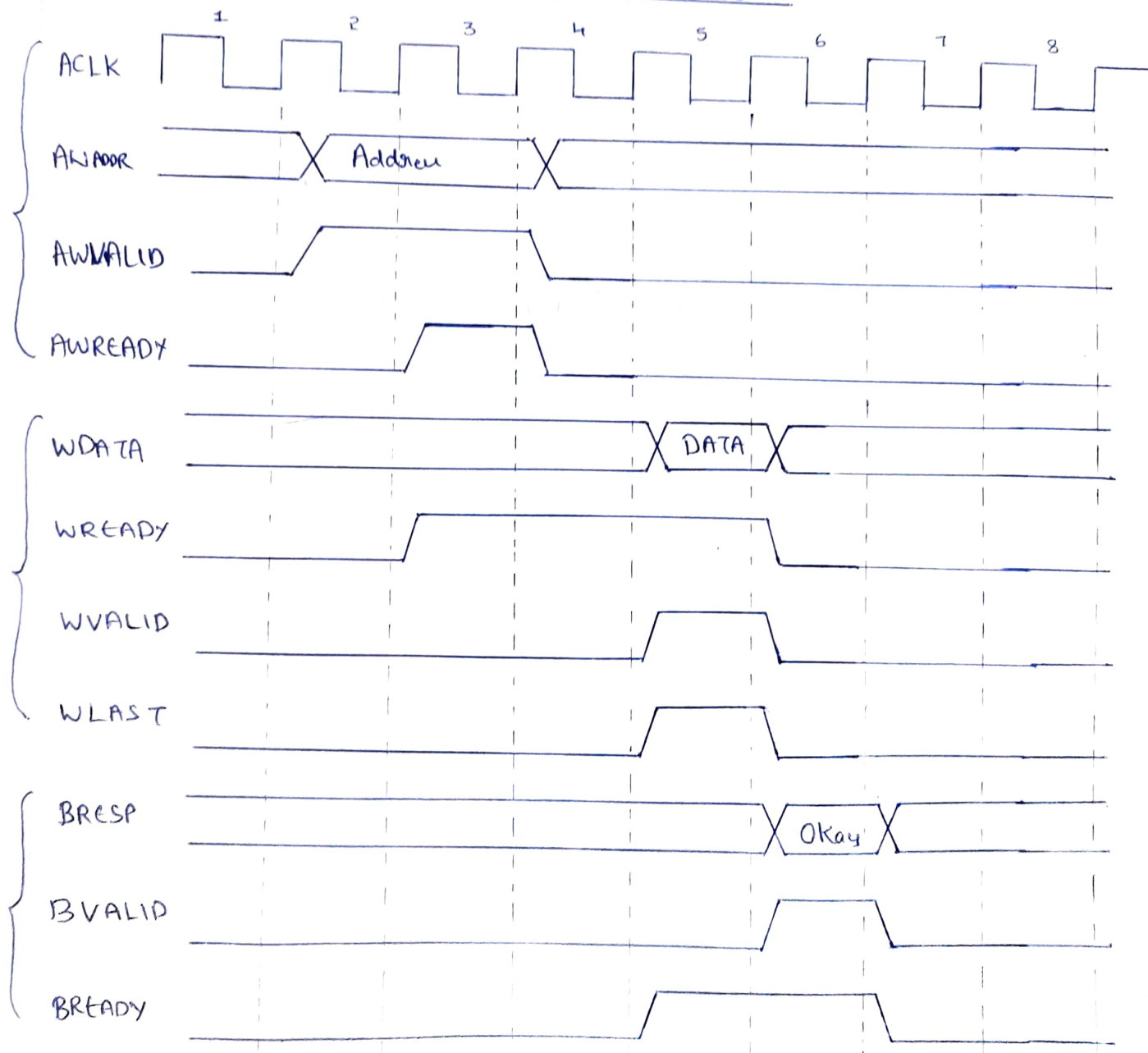
ARVALID	M	valid
ARREADY	S	Ready
ARID	M	ID
ARADDR	M	Address
ARLEN	M	Burst length
ARSIZE	M	Transaction Size
ARBURST	M	Transaction type

5. Read data

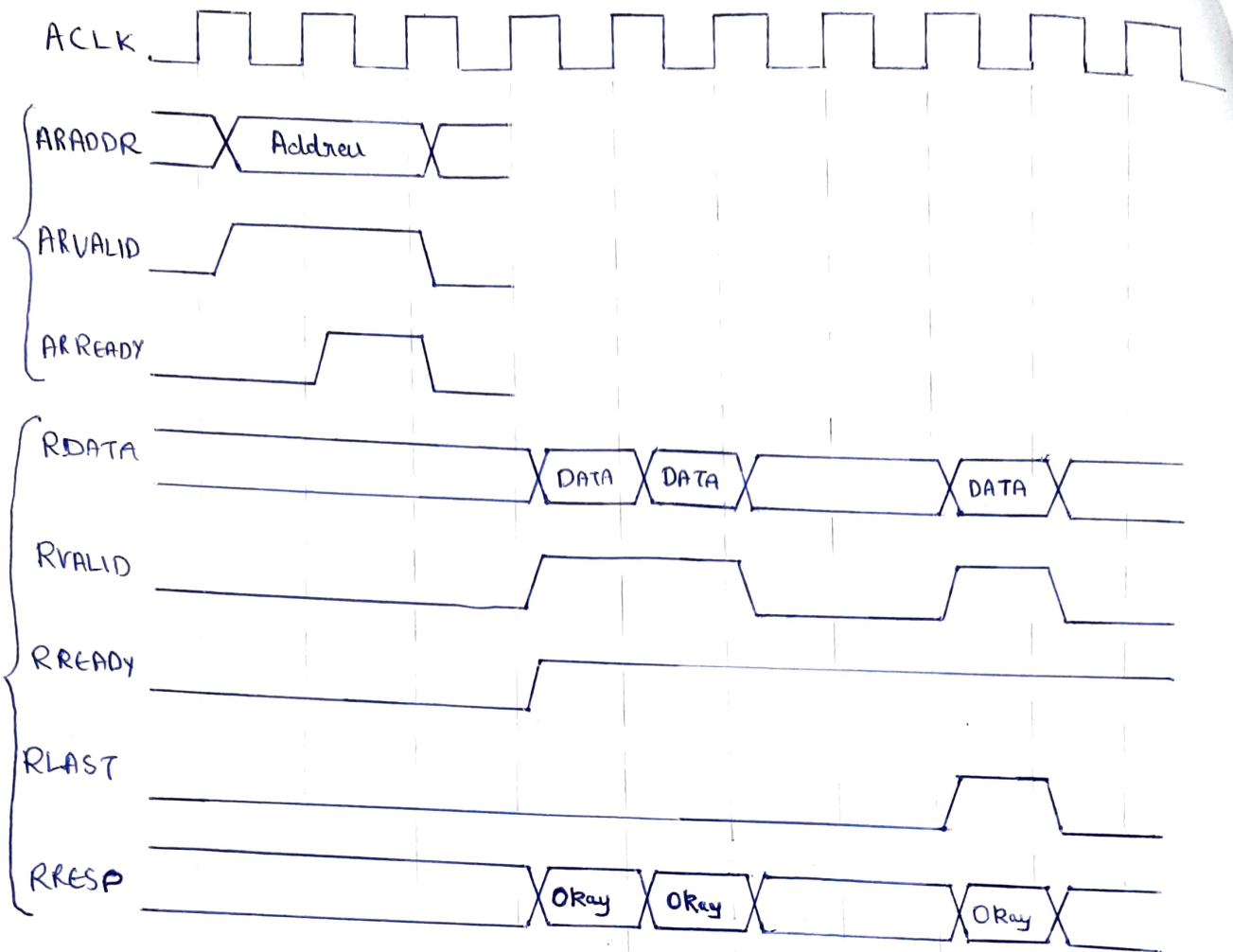
RVALID	S	Valid
RREADY	M	Ready
RID	S	Identifier
RDATA	S	Read data
RRESP	S	Read Response
RLAST	S	Last read data

Timing Diagram

Write transaction



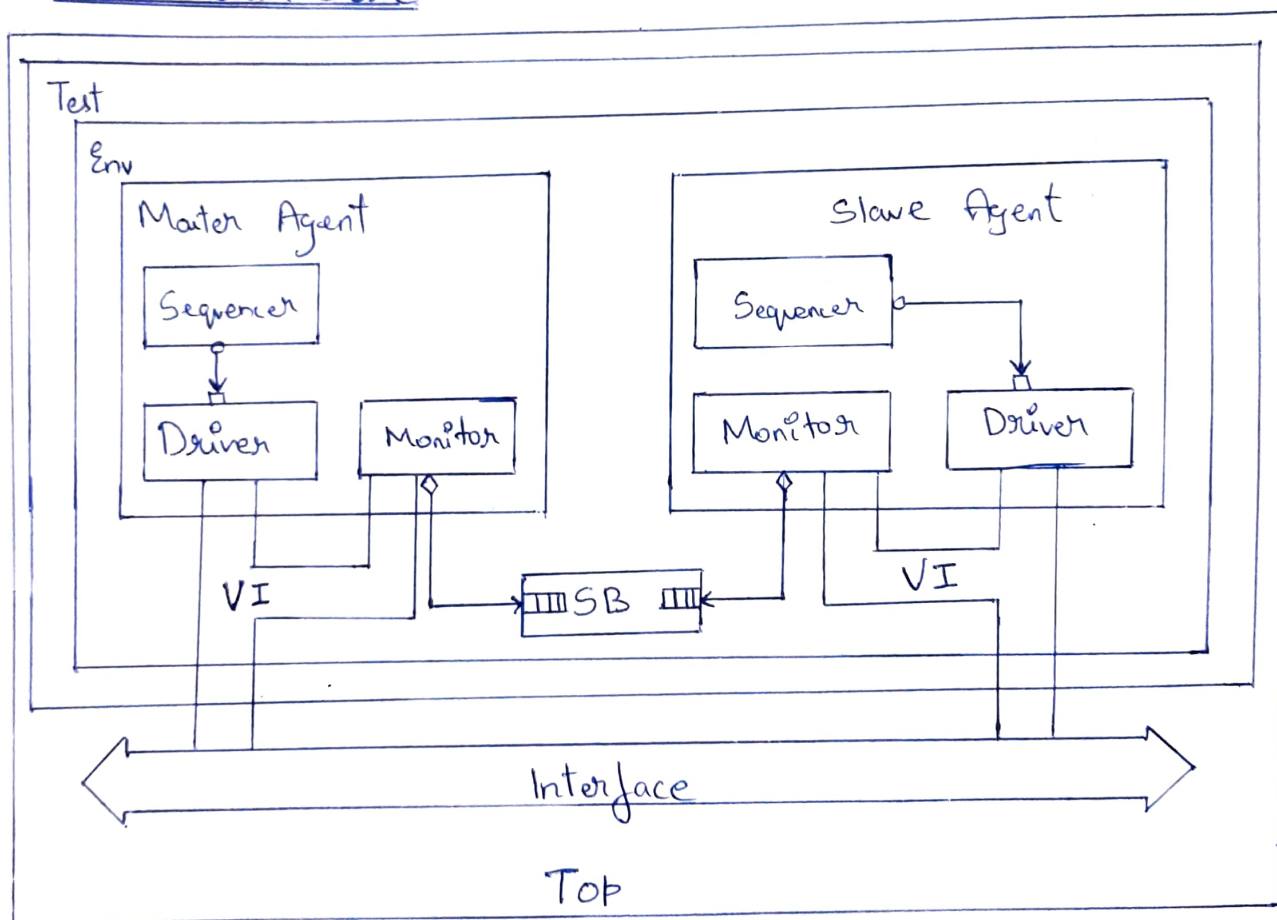
Read Transaction



Features of AXI Protocol

- 1) Burst based Communication. (Fixed, Increment and Wrapping)
- 2) Multiple Outstanding Requests
- 3) Independent read and write Channels
- 4) No strict timing relationship b/w address and data operations
- 5) Support for unaligned data transfer
- 6) Out of Order Completion

TB Architecture



Test Cases

- 1) Fixed :- In this type of transaction, the same location is written. Therefore, the start address is the address for all transfers of the transaction. $AWBURST$ or $ARBURST = 0$
Testing with varied length and size of transfer
- 2) Increment :- $AXBURST = 1$. Should check whether sequential memory access is made or not
- 3) Wrap :- $AXBURST = 2$. Should check if the transfer wraps when $\text{Address-N} = \text{wrap-boundary} + \text{Size} * \text{Burst-length}$.

Coverage Model

- 1) $AXBURST = \{0, 1, 2\}$
- 2) $AWLEN = ARLEN = \{1, 3, 7\}$
- 3) $AXSIZE = \{\cancel{2}, \cancel{4}, \cancel{8}\} \{1, 2, 4\}$
- 4) $RRESP = BRESP = \{0\}$

Assertions

- 1) The slave must wait for $AWVALID$, $AWREADY$, $WVALID$ and $WREADY$ to be asserted before asserting $BVALID$
- 2) The slave must wait for the last write data transfer before asserting $BVALID$
- 3) The slave must wait for both $ARVALID$ and $ARREADY$ before asserting $RVALID$ to indicate that valid data is available